

ALTERNATING CURRENTS

An understanding of the practical and economic advantages of transmission of power by electricity from Cambridge IGCSE / O Level Physics or equivalent is assumed.

21.1 Characteristics of alternating currents

Candidates should be able to:

- 1 understand and use the terms period, frequency and peak value as applied to an alternating current or voltage
- 2 use equations of the form $x = x_0 \sin \omega t$ representing a sinusoidally alternating current or voltage
- 3 recall and use the fact that the mean power in a resistive load is half the maximum power for a sinusoidal alternating current
- 4 distinguish between root-mean-square (r.m.s.) and peak values and recall and use $I_{\text{r.m.s.}} = I_0 / \sqrt{2}$ and $V_{\text{r.m.s.}} = V_0 / \sqrt{2}$ for a sinusoidal alternating current

1. O/N/09/41/Q7

A sinusoidal alternating voltage is to be rectified.

- (a) Suggest one advantage of full-wave rectification as compared with half-wave rectification.

.....
..... [1]

- (b) The rectification is produced using the circuit of Fig. 7.1.

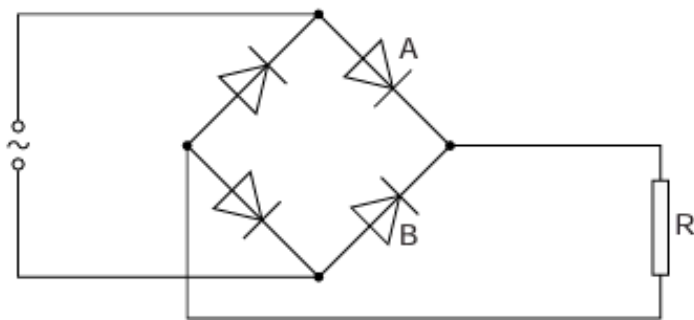


Fig. 7.1

All the diodes may be considered to be ideal.

The variation with time t of the alternating voltage applied to the circuit is shown in Fig. 7.2 and in Fig. 7.3.

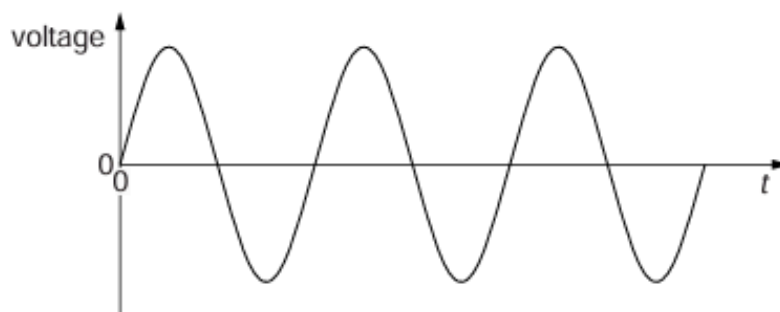
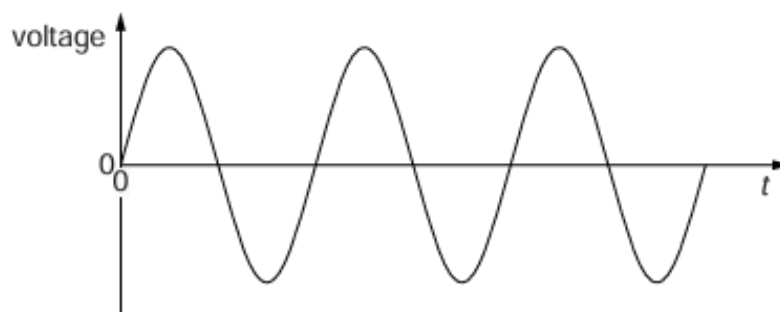


Fig. 7.2



- (i) On the axes of Fig. 7.2, draw a graph to show the variation with time t of the potential difference across diode A. [1]
- (ii) On the axes of Fig. 7.3, draw a graph to show the variation with time t of the potential difference across diode B. [1]
- (c) (i) On Fig. 7.1, draw the symbol for a capacitor, connected into the circuit so as to provide smoothing. [1]
- (ii) Fig. 7.4 shows the variation with time t of the smoothed potential difference across the resistor R in Fig. 7.1.

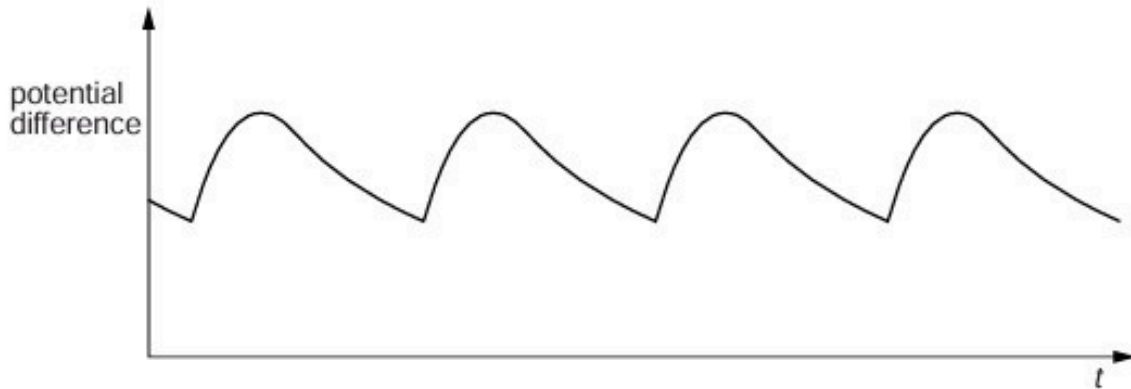


Fig. 7.4

1. State how the amount of smoothing may be increased.

.....
 [1]

2. On Fig. 7.4, draw the variation with time t of the potential difference across resistor R for increased smoothing. [2]

2. M/J/10/42/Q7

- (a) Explain what is meant by the *root-mean-square* (r.m.s.) value of an alternating voltage.

.....
.....
..... [2]

- (b) An alternating voltage V is represented by the equation

$$V = 220 \sin(120\pi t),$$

where V is measured in volts and t is in seconds.

For this alternating voltage, determine

- (i) the peak voltage,

peak voltage = V [1]

- (ii) the r.m.s. voltage,

r.m.s. voltage = V [1]

- (iii) the frequency.

frequency = Hz [1]

- (c) The alternating voltage in (b) is applied across a resistor such that the mean power output from the resistor is 1.5 kW.

Calculate the resistance of the resistor.

resistance = Ω [2]

3. O/N/10/43/Q6

The variation with time t of the current I in a resistor is shown in Fig. 6.1.

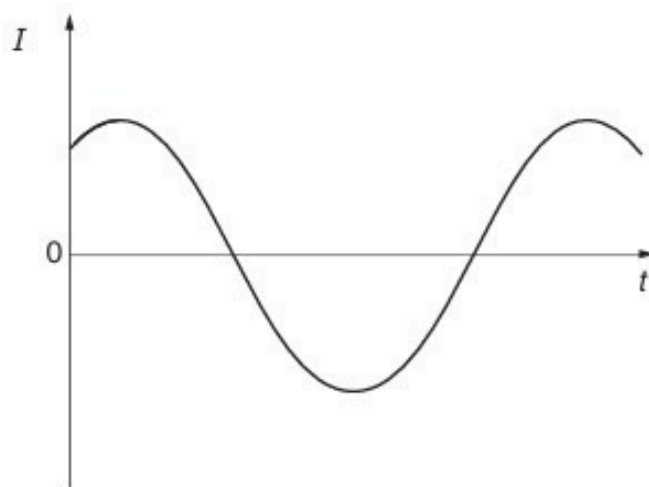


Fig. 6.1

The variation of the current with time is sinusoidal.

- (a) Explain why, although the current is not in one direction only, power is converted in the resistor.

.....
.....
..... [2]

- (b) Using the relation between root-mean-square (r.m.s.) current and peak current, deduce the value of the ratio

$$\frac{\text{average power converted in the resistor}}{\text{maximum power converted in the resistor}}.$$

ratio = [3]

4. M/J/11/42/Q6

An alternating current supply is connected in series with a resistor R , as shown in Fig. 6.1.

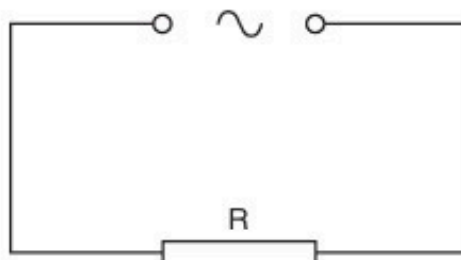


Fig. 6.1

The variation with time t (measured in seconds) of the current I (measured in amps) in the resistor is given by the expression

$$I = 9.9 \sin(380t).$$

(a) For the current in the resistor R , determine

(i) the frequency,

frequency = Hz [2]

(ii) the r.m.s. current.

r.m.s. current = A [2]

- (b) To prevent over-heating, the mean power dissipated in resistor R must not exceed 400W.
Calculate the minimum resistance of R.

resistance = Ω [2]

5. O/N/11/41/Q6

The variation with time t of the output V of an alternating voltage supply of frequency 50 Hz is shown in Fig. 6.1.

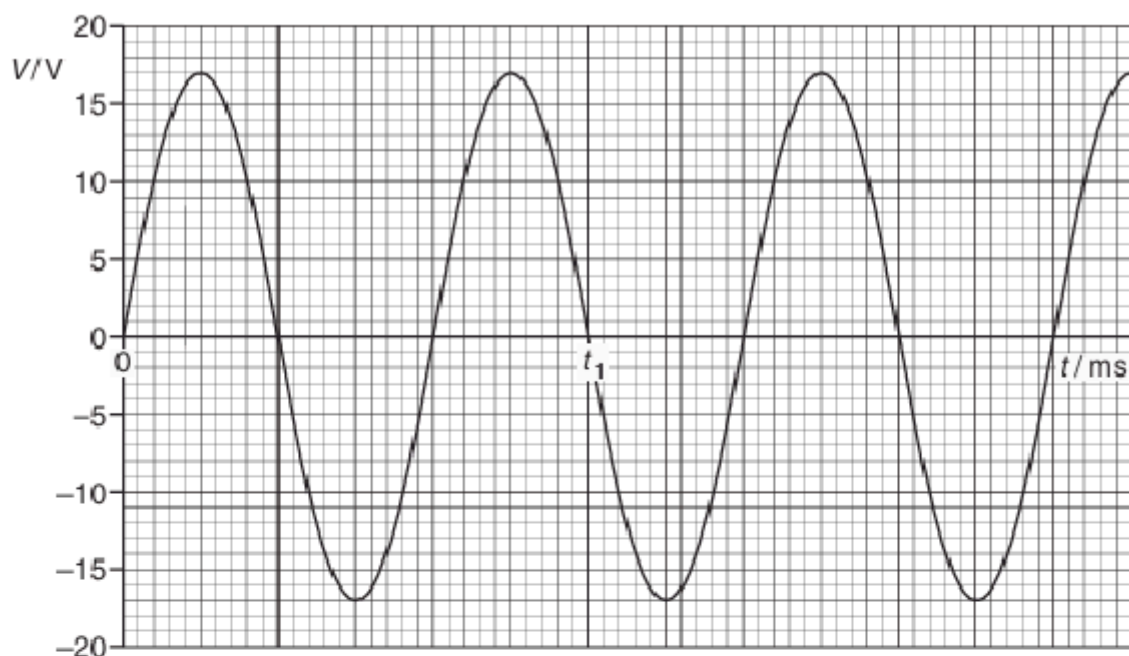


Fig. 6.1

(a) Use Fig. 6.1 to state

(i) the time t_1 ,

$$t_1 = \dots\dots\dots \text{ s [2]}$$

(ii) the peak value V_0 of the voltage,

$$V_0 = \dots\dots\dots \text{ V [1]}$$

(iii) the root-mean-square voltage V_{rms} ,

$$V_{\text{rms}} = \dots\dots\dots \text{ V [1]}$$

(iv) the mean voltage $\langle V \rangle$.

$$\langle V \rangle = \dots\dots\dots \text{ V [1]}$$

- (b) The alternating supply is connected in series with a resistor of resistance $2.4\ \Omega$. Calculate the mean power dissipated in the resistor.

power = W [2]

6. O/N/11/43/Q5

The components for a bridge rectifier are shown in Fig. 5.1.

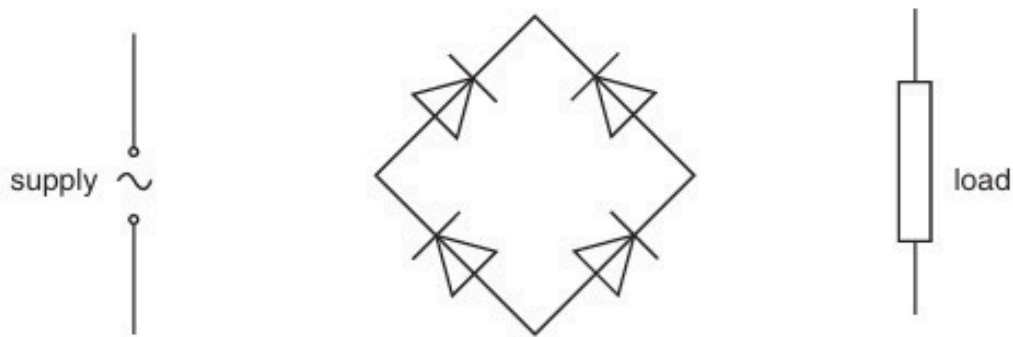


Fig. 5.1

- (a) Complete the circuit of Fig. 5.1 by showing the connections of the supply and of the load to the diodes. [2]

- (b) Suggest one advantage of the use of a bridge rectifier, rather than a single diode, for the rectification of alternating current.

.....
..... [1]

- (c) State

- (i) what is meant by *smoothing*,

.....
..... [1]

- (ii) the effect of the value of the capacitance of the smoothing capacitor in relation to smoothing.

.....
.....
..... [2]

7. M/J/12/41/Q6

A sinusoidal alternating voltage supply is connected to a bridge rectifier consisting of four ideal diodes. The output of the rectifier is connected to a resistor R and a capacitor C as shown in Fig. 6.1.

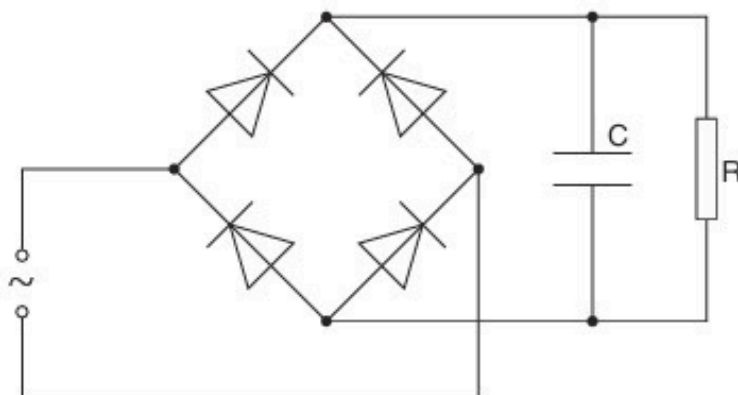


Fig. 6.1

The function of C is to provide some smoothing to the potential difference across R . The variation with time t of the potential difference V across the resistor R is shown in Fig. 6.2.

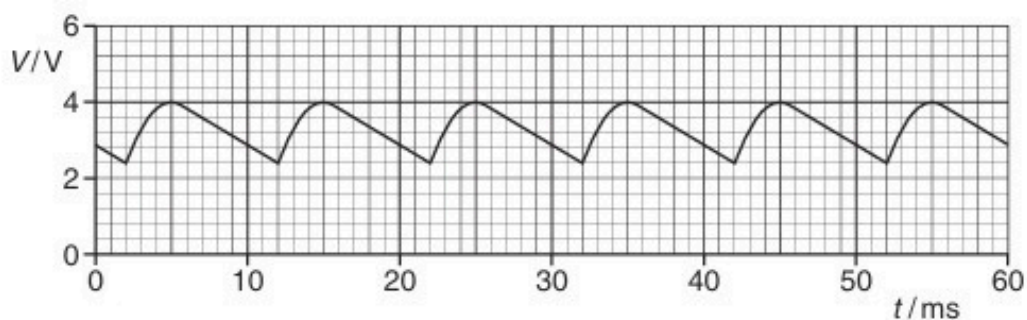


Fig. 6.2

(a) Use Fig. 6.2 to determine, for the alternating supply,

(i) the peak voltage,

peak voltage = V [1]

(ii) the root-mean-square (r.m.s.) voltage,

r.m.s. voltage = V [1]

(iii) the frequency. Show your working.

frequency = Hz [2]

(b) The capacitor C has capacitance $5.0\ \mu\text{F}$.

For a single discharge of the capacitor through the resistor R, use Fig. 6.2 to

(i) determine the change in potential difference,

change = V [1]

(ii) determine the change in charge on each plate of the capacitor,

change = C [2]

(iii) show that the average current in the resistor is $1.1 \times 10^{-3}\text{ A}$.

- (c) Use Fig. 6.2 and the value of the current given in (b)(iii) to estimate the resistance of resistor R.

resistance = Ω [2]

8. O/N/12/43/Q6

A bridge rectifier consists of four ideal diodes A, B, C and D, connected as shown in Fig. 6.1.

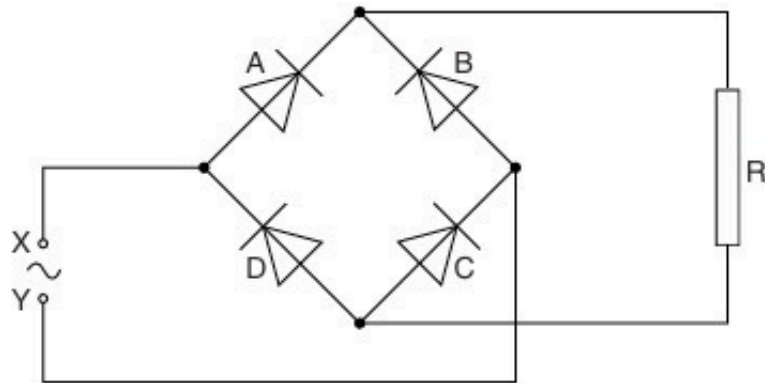


Fig. 6.1

An alternating supply is applied between the terminals X and Y.

(a) (i) On Fig. 6.1, label the positive (+) connection to the load resistor R. [1]

(ii) State which diodes are conducting when terminal Y of the supply is positive.

diode and diode [1]

(b) The variation with time t of the potential difference V across the load resistor R is shown in Fig. 6.2.

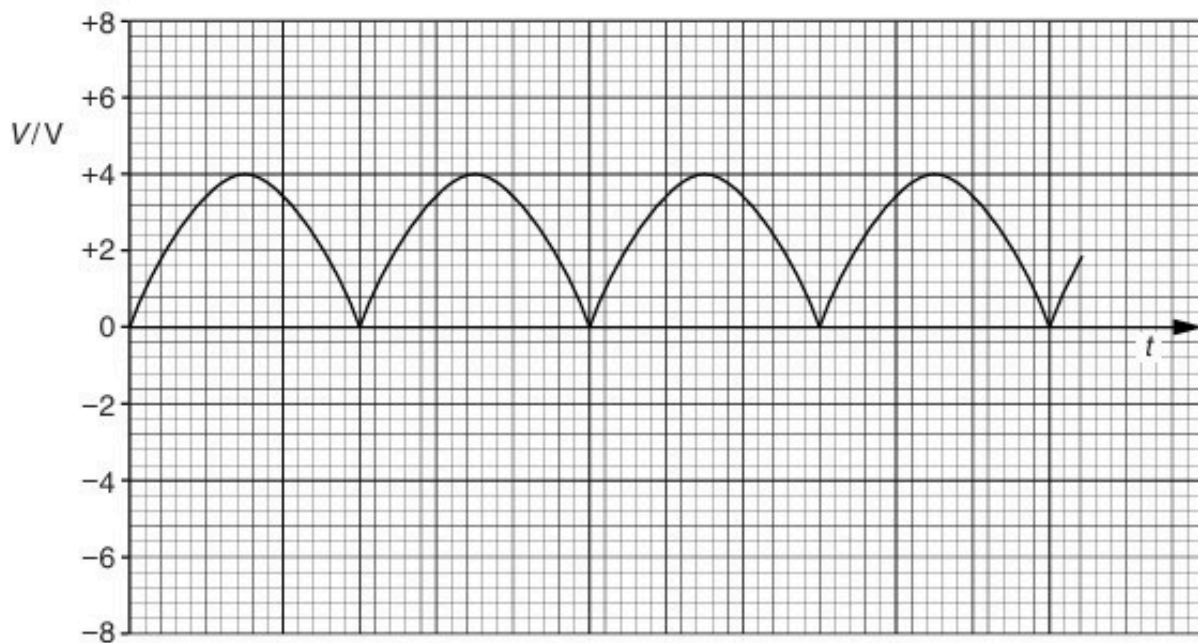


Fig. 6.2

The load resistor R has resistance $2700\ \Omega$.

- (i) Use Fig. 6.2 to determine the mean power dissipated in the resistor R.

power = W [3]

- (ii) On Fig. 6.1, draw the symbol for a capacitor, connected so as to increase the mean power dissipated in the resistor R. [1]

- (c) The capacitor in (b)(ii) is now removed from the circuit.
The diode A in Fig. 6.1 stops functioning, so that it now has infinite resistance.

On Fig. 6.2, draw the variation with time t of the new potential difference across the resistor R. [2]

9. O/N/15/43/Q7

A student is using a power supply that produces a sinusoidal output. The meters on the supply show that the output voltage V has a root-mean-square (r.m.s.) value of 14V with a frequency of 750 Hz.

The variation with time t of the output voltage V may be represented by the expression

$$V = V_0 \sin \omega t.$$

(a) Determine the value of

(i) V_0 ,

$$V_0 = \dots\dots\dots \text{ V [1]}$$

(ii) ω .

$$\omega = \dots\dots\dots \text{ rad s}^{-1} \text{ [1]}$$

(b) A capacitor with a large capacitance is connected across the terminals of the supply.

Suggest and explain why this may lead to a large current from the supply.

.....
.....
..... [3]

10. M/J/16/41/Q11

The variation with time t of the sinusoidal current I in a resistor of resistance $450\,\Omega$ is shown in Fig. 11.1.

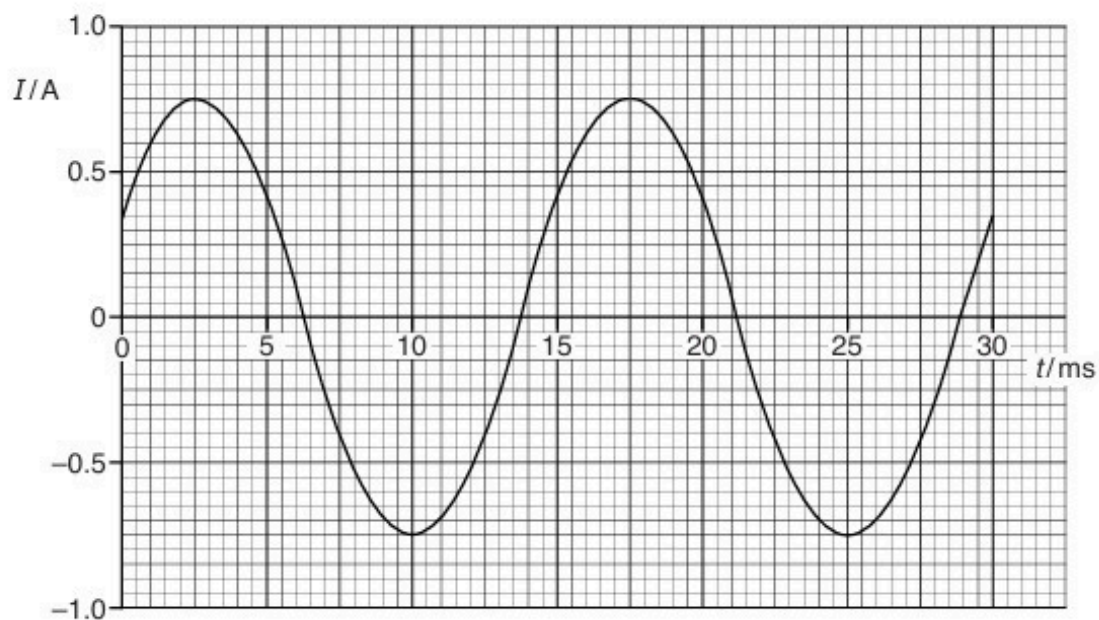


Fig. 11.1

Use data from Fig. 11.1 to determine, for the time $t = 0$ to $t = 30\text{ ms}$,

(a) the frequency of the current,

frequency = Hz [2]

(b) the mean current,

mean current = A [1]

(c) the root-mean-square (r.m.s.) current,

r.m.s. current = A [2]

(d) the energy dissipated by the resistor.

energy =J [2]

[Total: 7]

11. M/J/16/42/Q11

A bridge rectifier contains four ideal diodes A, B, C and D, as shown in Fig. 11.1.

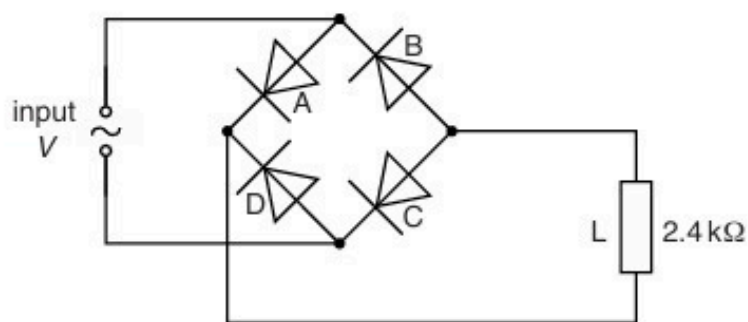


Fig. 11.1

The output of the rectifier is connected to a load L of resistance $2.4\text{ k}\Omega$.

- (a) On Fig. 11.1, mark with the letter P the positive terminal of the load. [1]
- (b) The variation with time t of the potential difference V across the input to the rectifier is shown in Fig. 11.2.

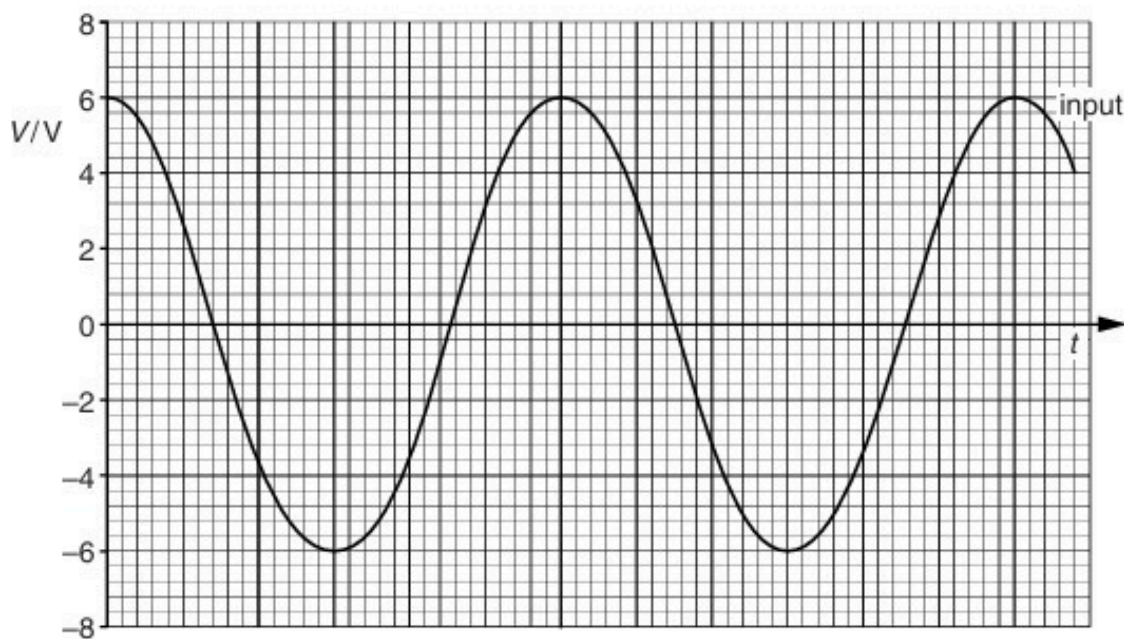


Fig. 11.2

Calculate the root-mean-square (r.m.s.) current in the load L .

(c) The potential difference across the load L is to be smoothed using a capacitor.

(i) On Fig. 11.1, draw the symbol for a capacitor, connected to produce smoothing. [1]

(ii) The minimum potential difference across the load L with the smoothing capacitor connected is 3.0 V.

On Fig. 11.2, sketch the variation with time t of the potential difference across the load L . [3]

[Total: 7]

12. O/N/17/42/Q11

The circuit for a full-wave rectifier using four ideal diodes is shown in Fig. 11.1.

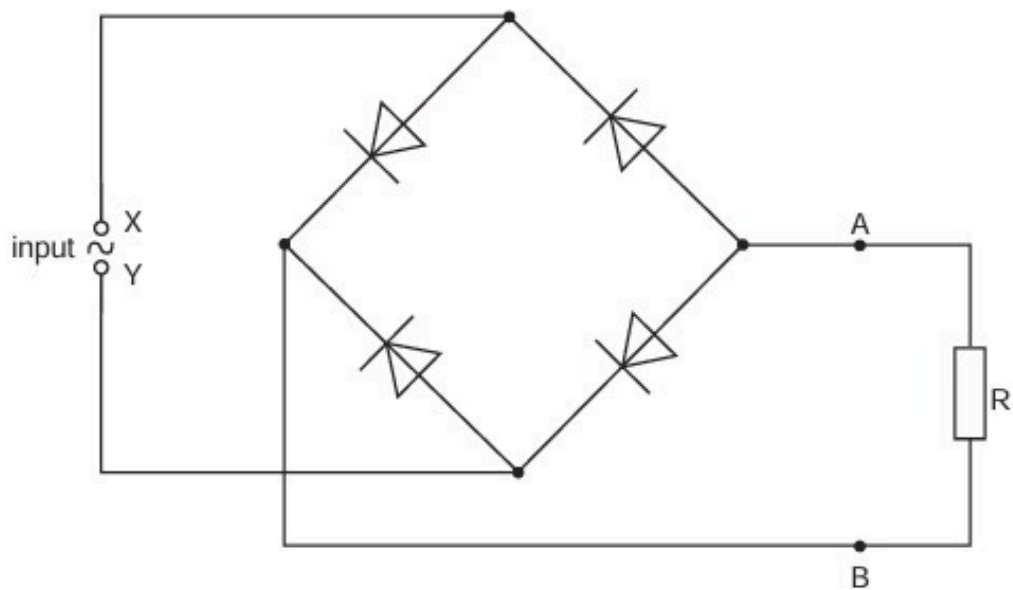


Fig. 11.1

A resistor R is connected across the output AB of the rectifier.

(a) On Fig. 11.1,

- (i)** draw a circle around any diodes that conduct when the terminal X of the input is positive with respect to terminal Y, [1]
- (ii)** label the positive (+) and the negative (–) terminals of the output AB. [1]

- (b) The variation with time t of the potential difference V across the input XY is given by the expression

$$V = 5.6 \sin 380t$$

where V is measured in volts and t is measured in seconds.

The variation with time t of the rectified potential difference across the resistor R is shown in Fig. 11.2.

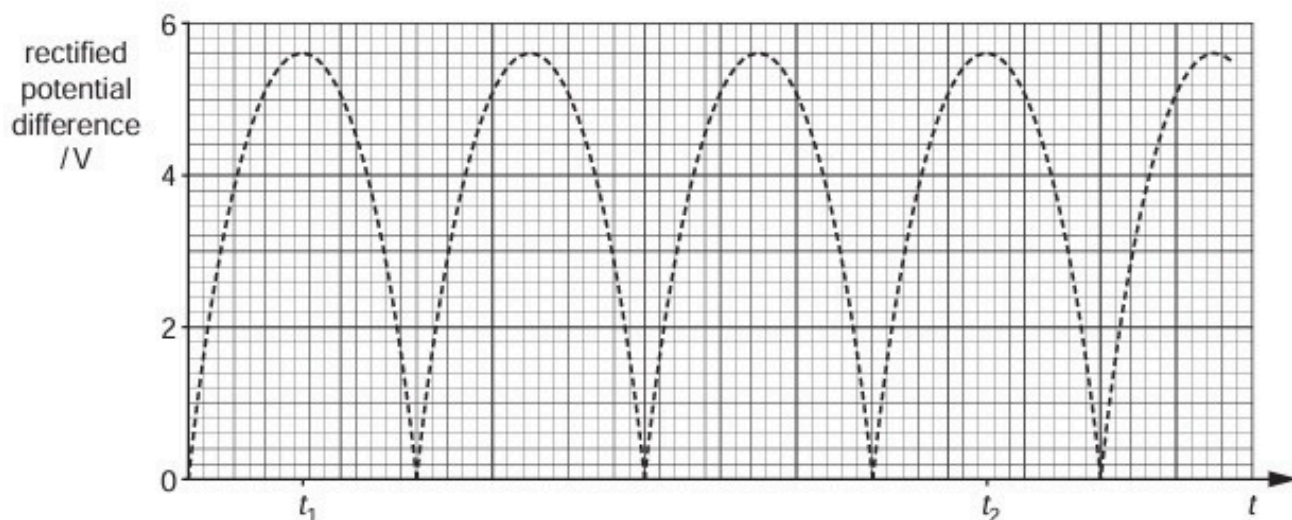


Fig. 11.2

Use the expression for the input potential difference V , or otherwise, to determine

- (i) the root-mean-square (r.m.s.) potential difference $V_{\text{r.m.s.}}$ of the input,

$$V_{\text{r.m.s.}} = \dots\dots\dots \text{ V [1]}$$

- (ii) the number of times per second that the rectified potential difference at the output reaches a peak value.

number = [2]

- (c) A capacitor is now connected between the terminals AB of the output.
The capacitor reduces the variation (the ripple) in the output to 1.6 V.

- (i) On Fig. 11.2, sketch the variation with time t of the smoothed output voltage for time $t = t_1$ to time $t = t_2$. [4]

- (ii) Suggest and explain the effect, if any, on the mean power dissipation in resistor R when the capacitor is connected between terminals AB.

.....
.....
.....[2]

[Total: 11]

13. M/J/19/41/Q10

A bridge rectifier contains four diodes. The output of the rectifier is connected to a resistor R , as shown in Fig. 10.1.

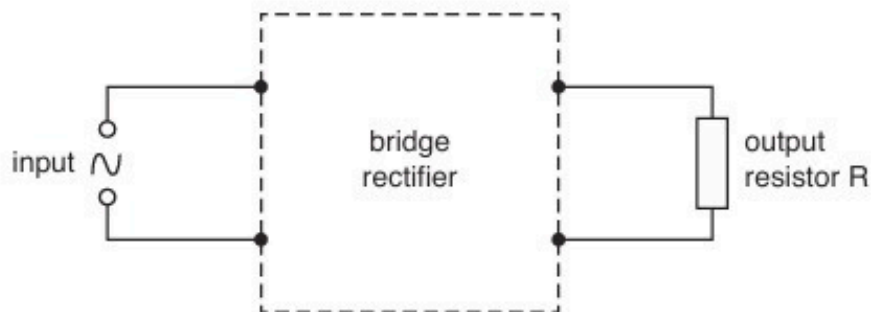


Fig. 10.1

The variation with time t of the input e.m.f. E to the rectifier is given by the expression

$$E = 15 \cos(210t)$$

where t is measured in seconds and E in volts.

The variation with time t of the potential difference V across resistor R is shown in Fig. 10.2.

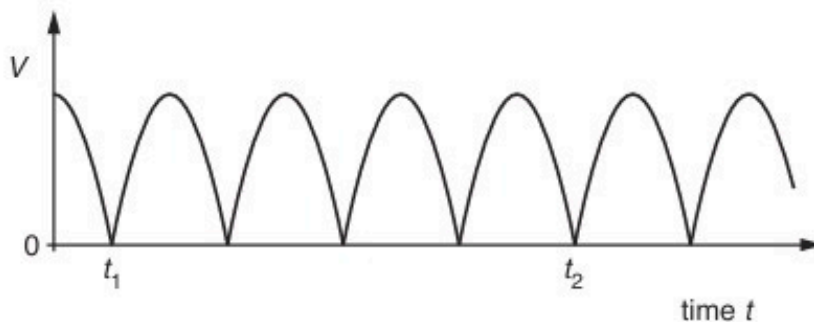


Fig. 10.2

Determine:

- (a) the maximum potential difference V_{MAX} across resistor R

$$V_{\text{MAX}} = \dots\dots\dots \text{V} \quad [1]$$

(b) the time interval, to two significant figures, between time t_1 and time t_2 .

time = s [3]

[Total: 4]

14. O/N/19/41/Q10

A bridge rectifier using four ideal diodes is shown in Fig. 10.1.

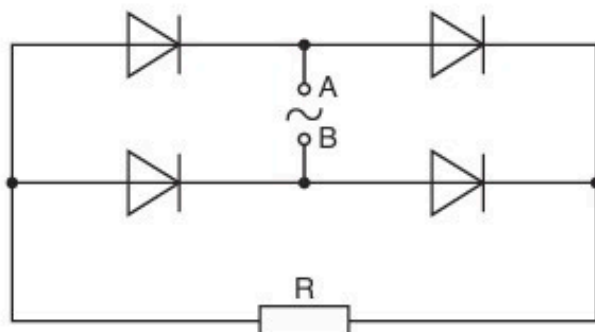


Fig. 10.1

The sinusoidal alternating electromotive force (e.m.f.) applied between points A and B has a root-mean-square (r.m.s.) value of 7.0 V.

- (a) (i) On Fig. 10.1, circle the diodes that conduct when point B is positive with respect to point A. [1]
- (ii) Calculate the maximum potential difference V_{MAX} across resistor R.

$V_{\text{MAX}} = \dots\dots\dots \text{ V [1]}$

- (b) A capacitor is connected into the circuit to produce smoothing of the potential difference across resistor R.

The variation with time t of the potential difference V across resistor R is shown in Fig. 10.2.

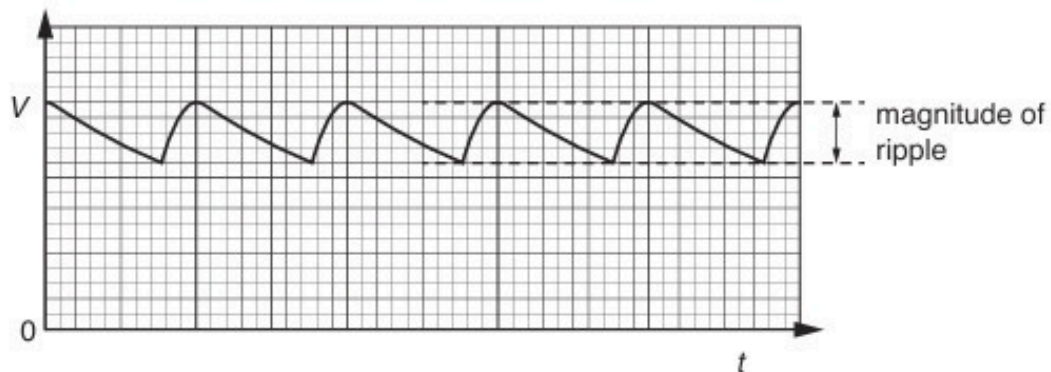


Fig. 10.2

- (i) On Fig. 10.1, draw the symbol for a capacitor, connected so as to produce smoothing. [1]

- (ii) State the effect, if any, on the magnitude of the ripple on V when, separately:

1. a capacitor of larger capacitance is used

.....

2. the resistor R has a smaller resistance.

.....

[2]

[Total: 5]

15. F/M/21/42/Q10

The output potential difference (p.d.) of an alternating power supply is represented by

$$V = 320 \sin(100\pi t)$$

where V is the p.d. in volts and t is the time in seconds.

- (a) Determine the root-mean-square (r.m.s.) p.d. of the power supply.

r.m.s. p.d. = V [1]

- (b) Determine the period T of the output.

$T =$ s [2]

- (c) The power supply is connected to resistor R and a diode in the circuit shown in Fig. 10.1.

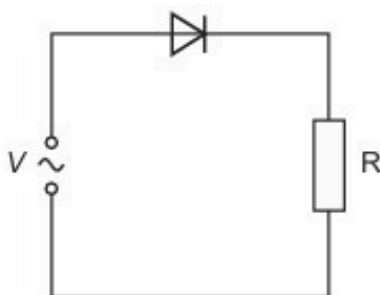


Fig. 10.1

- (i) State the name of the type of rectification produced by the diode in Fig. 10.1.

..... [1]

- (ii) On Fig. 10.2 sketch the variation with time t of the p.d. V_R across R from time $t = 0$ to time $t = 40$ ms.

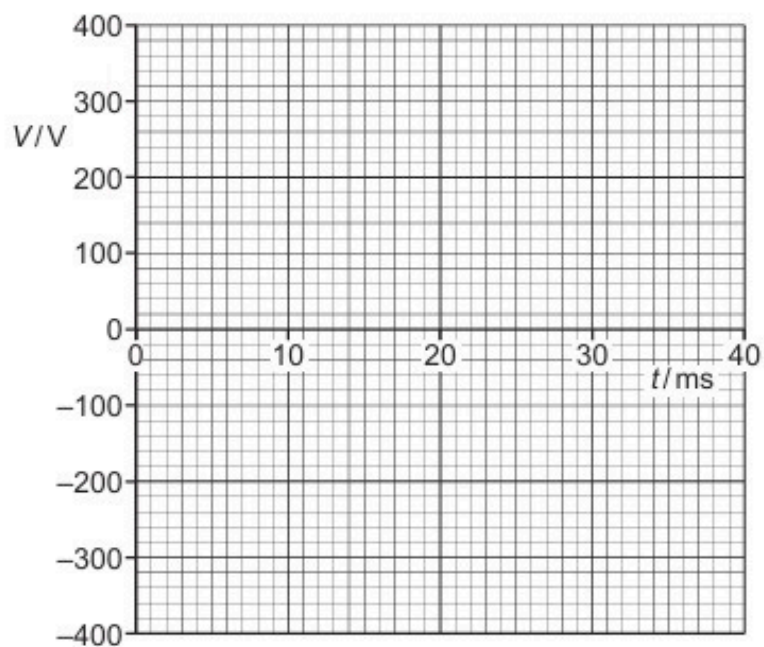


Fig. 10.2

[3]

- (iii) On Fig. 10.1, draw the symbol for a component that may be connected to produce smoothing of V_R .

[1]

[Total: 8]

16. M/J/21/42/Q10

- (a) By reference to heating effect, explain what is meant by the *root-mean-square (r.m.s.)* value of an alternating current.

.....
.....
..... [2]

- (b) The variations with time t of two currents I_1 and I_2 are shown in Fig. 10.1 and Fig. 10.2.

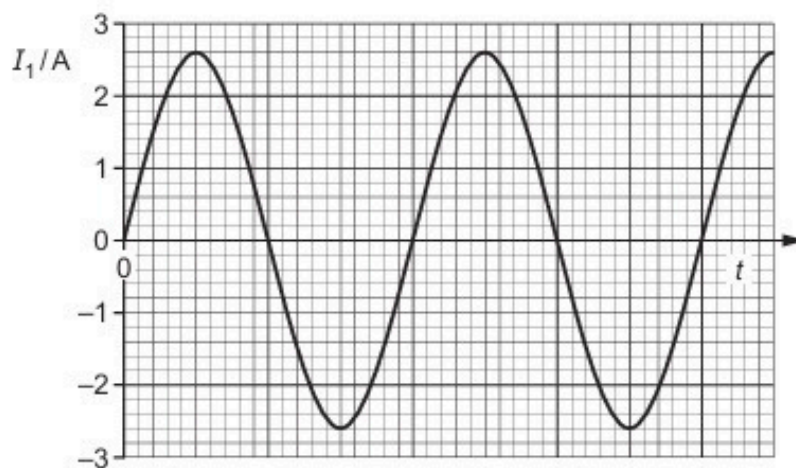


Fig. 10.1

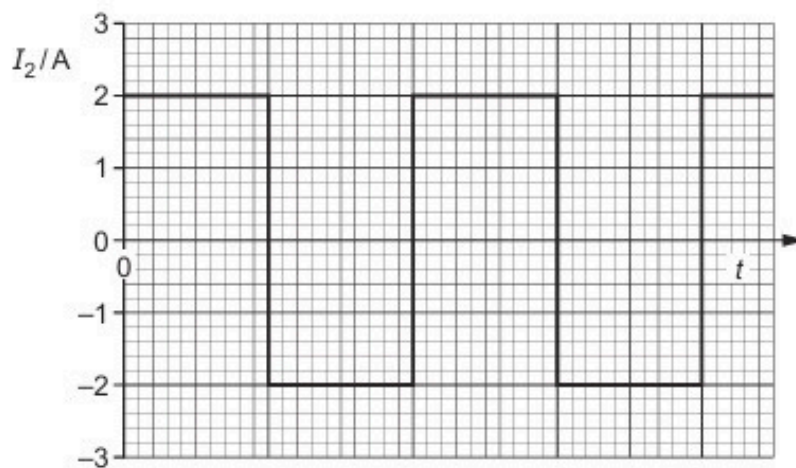


Fig. 10.2

- (i) Use Fig. 10.1 to determine the peak value and the r.m.s. value of the current I_1 .

peak value = A

r.m.s. value = A
[1]

- (ii) Use Fig. 10.2 to determine the peak value and the r.m.s. value of the current I_2 .

peak value = A

r.m.s. value = A
[1]

- (c) The variation with time t of the supply voltage V to a house is given by the expression

$$V = 240 \sin kt$$

where V is in volts, t is in seconds and k is a constant with unit rad s^{-1} .

- (i) The frequency of the supply voltage is 50 Hz.

Determine k to two significant figures.

$k = \dots\dots\dots \text{rad s}^{-1}$ [2]

- (ii) The supply voltage is applied to a heater. The mean power of the heater is 3.2 kW.

Calculate the resistance of the heater.

resistance = Ω [2]

[Total: 8]

17. O/N/21/41/Q9

- (a) State, by reference to the power dissipated in a resistor, what is meant by the *root-mean-square (r.m.s.)* value of an alternating voltage.

.....

 [2]

- (b) A coil is rotating freely, on frictionless bearings, at constant speed in a uniform magnetic field. This rotation causes an induced alternating electromotive force (e.m.f.) across the open terminals of the coil. The induced e.m.f. has r.m.s. value 12 V and frequency 50 Hz.

The speed of rotation of the coil is now doubled.

- (i) State and explain, with reference to the principles of electromagnetic induction, the effect of the increased speed of rotation on the r.m.s. value of the induced e.m.f.

.....

 [2]

- (ii) On Fig. 9.1, sketch the variation with time t of the induced e.m.f. E across the terminals of the coil at the **increased** speed of rotation. Your line should extend from time $t = 0$ to time $t = 20$ ms. Assume that $E = 0$ when $t = 0$.

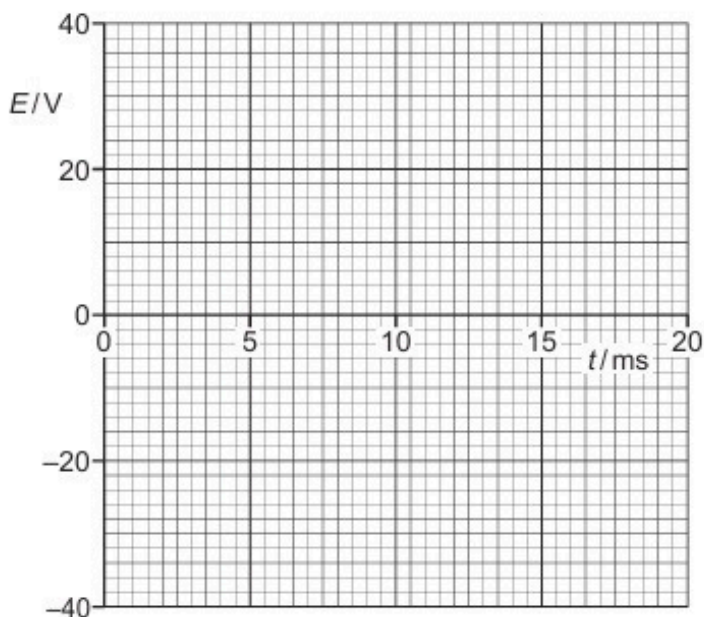


Fig. 9.1

[3]

- (c) State and explain the effect on the motion of the coil in (b) of connecting a load resistor across its terminals.

.....

.....

.....

..... [2]

[Total: 9]

18. F/M/22/42/Q7

- (a) Alternating current (a.c.) is converted into direct current (d.c.) using a full-wave rectification circuit. Part of the diagram of this circuit is shown in Fig. 7.1.

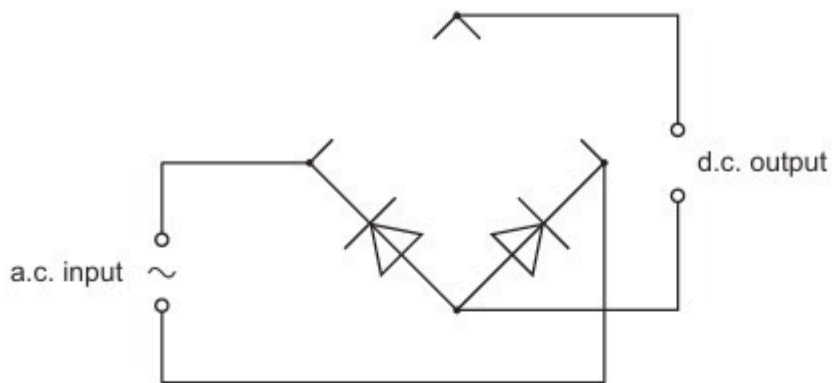


Fig. 7.1

- (i) Complete the circuit in Fig. 7.1 by adding the necessary components in the gaps. [1]
- (ii) On Fig. 7.1 mark with a + the positive output terminal of the rectifier. [1]
- (b) The output voltage V of an a.c. power supply varies sinusoidally with time t as shown in Fig. 7.2.

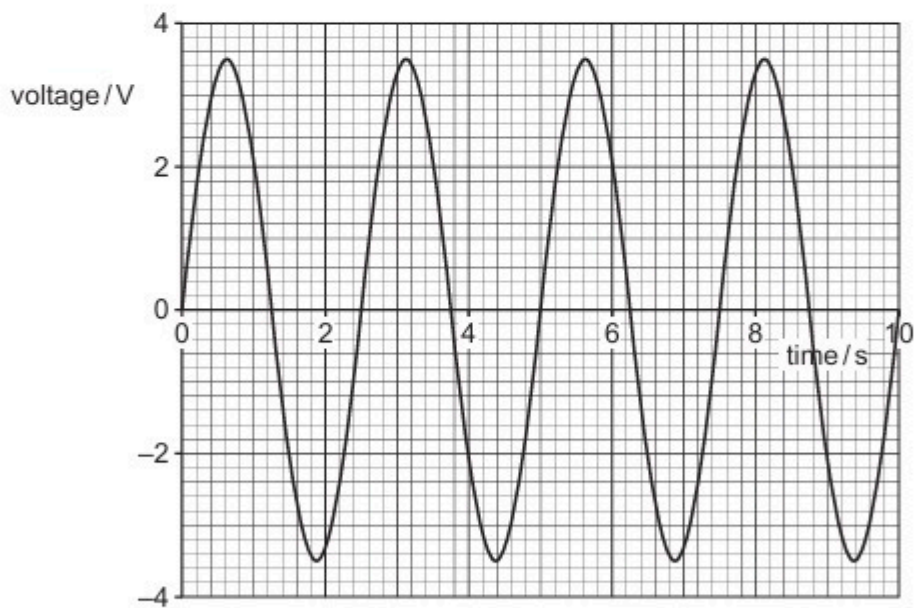


Fig. 7.2

- (i) Determine the equation for V in terms of t , where V is in volts and t is in seconds.

$V = \dots\dots\dots$ [2]

- (ii) The supply is connected to a $12\ \Omega$ resistor. Calculate the mean power dissipated in the resistor.

mean power = W [2]

[Total: 6]

19. M/J/22/41/Q5

Fig. 5.1 shows four diodes and a load resistor of resistance $1.2\text{ k}\Omega$, connected in a circuit that is used to produce rectification of an alternating voltage.

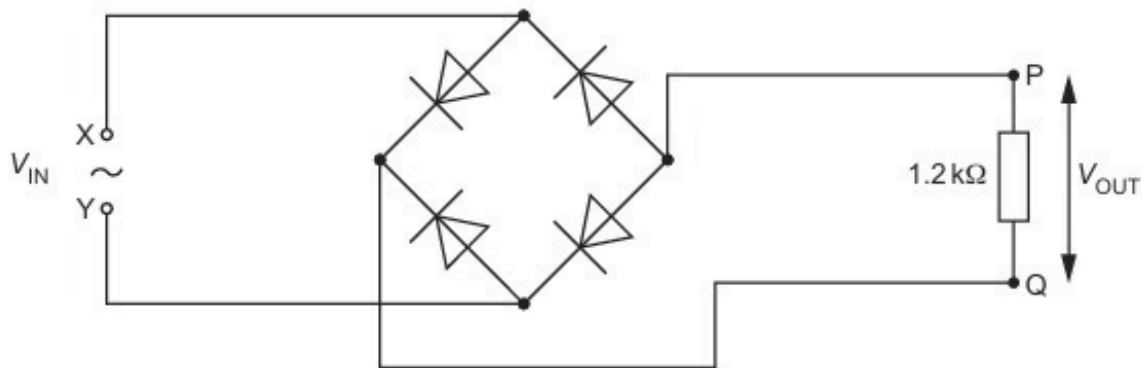


Fig. 5.1

- (a) (i) State what is meant by rectification.

.....
..... [1]

- (ii) State the type of rectification produced by the circuit in Fig. 5.1.

..... [1]

- (b) A sinusoidal alternating voltage V_{IN} is applied across the input terminals X and Y. The variation with time t of V_{IN} is given by the equation

$$V_{IN} = 6.0 \sin 25\pi t$$

where V_{IN} is in volts and t is in seconds.

- (i) On Fig. 5.1, label the output terminals P and Q with the appropriate symbols to indicate the polarity of the output voltage V_{OUT} . [1]

- (ii) The magnitude of the output voltage V_{OUT} varies with t as shown in Fig. 5.2.

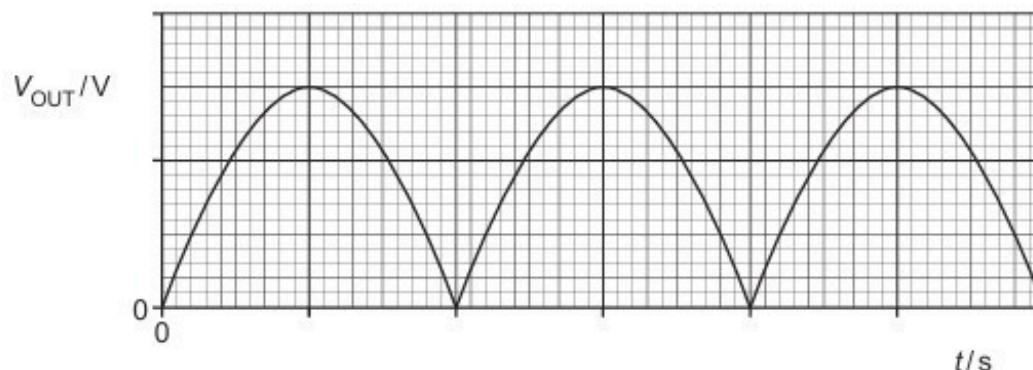


Fig. 5.2

On Fig. 5.2, label both of the axes with the correct scales. Use the space below for any working that you need.

[3]

- (c) The output voltage in (b) is smoothed by adding a capacitor to the circuit in Fig. 5.1.
The difference between the maximum and minimum values of the smoothed output voltage is 10% of the peak voltage.

- (i) On Fig. 5.1, draw the circuit symbol for a capacitor showing the capacitor correctly connected into the circuit. [1]

- (ii) On Fig. 5.2, sketch the variation with t of the smoothed output voltage. [2]

- (iii) Calculate the capacitance C of the capacitor.

$C = \dots\dots\dots$ F [3]

[Total: 12]

20. O/N/22/41/Q7

- (a) A sinusoidal alternating voltage has a root-mean-square (r.m.s.) potential difference (p.d.) of 4.2 V and a frequency of 50 kHz.

- (i) The alternating voltage is applied across a resistor of resistance 760 Ω .

By considering the peak voltage, show that the maximum power dissipated by the resistor is 46 mW.

[2]

- (ii) On Fig. 7.1, draw a smooth curve to show how the power P dissipated in the resistor varies with time t between $t = 0$ and $t = 40 \mu\text{s}$. Assume that $P = 0$ when $t = 0$.

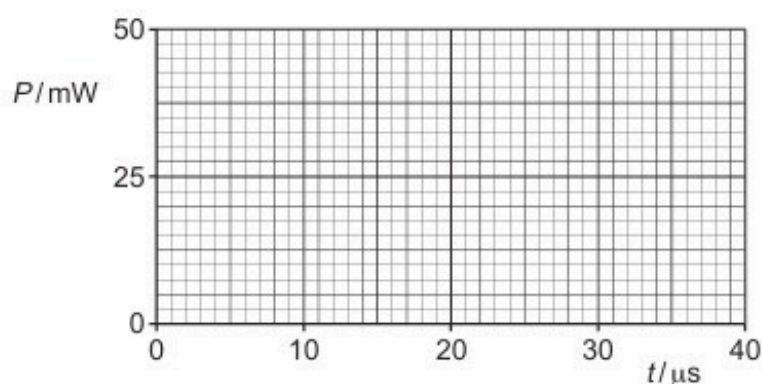


Fig. 7.1

[3]

- (iii) Use your line in (a)(ii) to explain why the mean power dissipated in the resistor is 23 mW.

.....

 [1]

(b) The alternating voltage in **(a)** is now applied to a piezoelectric crystal in air.

(i) Explain what happens to the air surrounding the crystal.

.....

.....

.....

..... [3]

(ii) A second piezoelectric crystal is placed in the air near to the first crystal.

Explain the effect of the surrounding air in **(b)(i)** on the second crystal.

..... [1]

[Total: 10]

21. M/J/23/41/Q5

Part of an electric circuit is shown in Fig. 5.1.

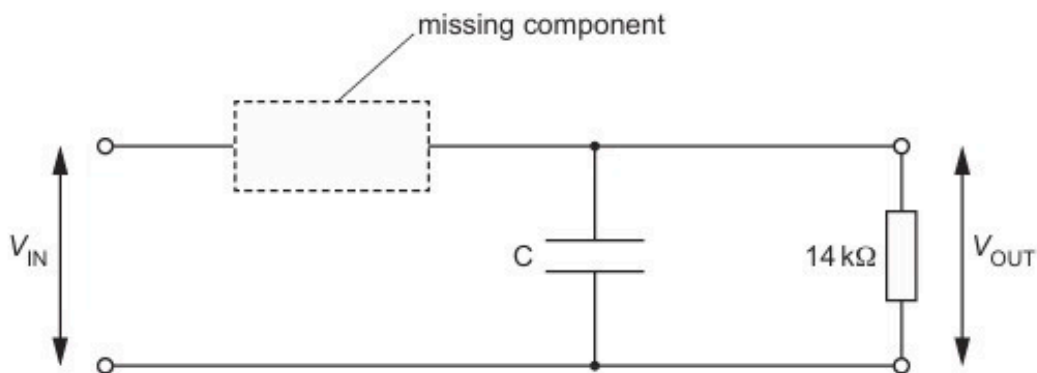


Fig. 5.1

The circuit is used to produce half-wave rectification of an alternating voltage of potential difference (p.d.) V_{IN} .

The output p.d. across the $14\text{ k}\Omega$ resistor is V_{OUT} .

- (a) (i)** A component is missing from the circuit of Fig. 5.1.

Complete the circuit diagram in Fig. 5.1 by adding the circuit symbol for the missing component, correctly connected. [1]

- (ii)** A capacitor C is shown in the circuit of Fig. 5.1.

State the effect on V_{OUT} of including the capacitor in the circuit.

..... [1]

- (b)** Fig. 5.2 shows the variation with time t of V_{IN} .

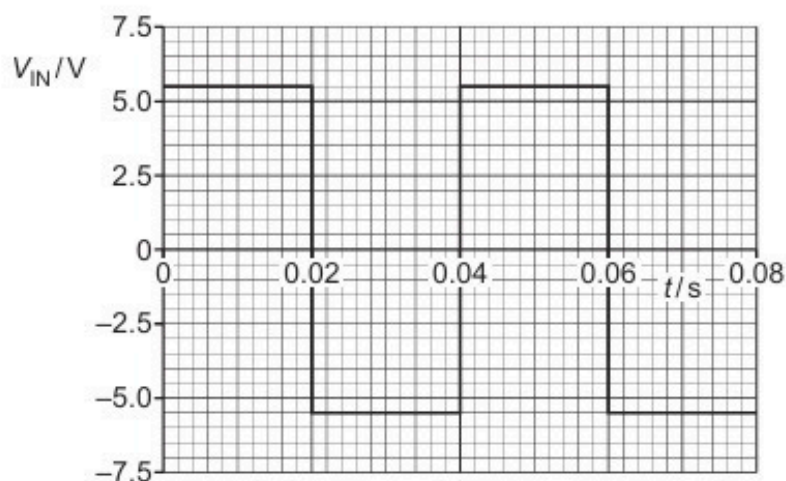


Fig. 5.2

Fig. 5.3 shows the variation with t of V_{OUT} .

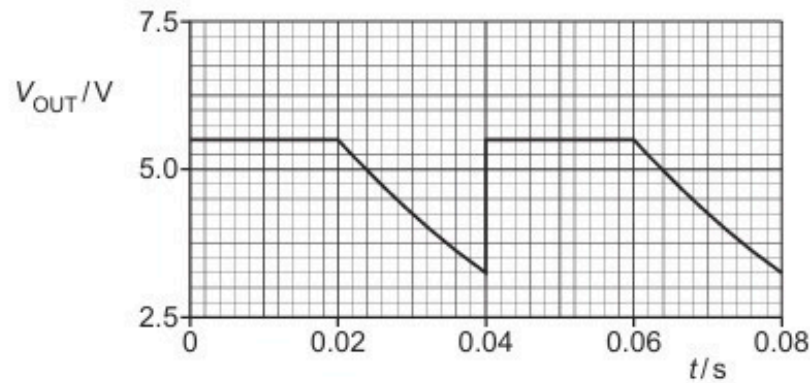


Fig. 5.3

- (i) Determine the frequency of V_{IN} .

frequency = Hz [1]

- (ii) Show that the time constant τ for the discharge of the capacitor through the resistor is 0.038 s.

[2]

- (iii) Calculate the capacitance of C. Give a unit with your answer.

capacitance = unit [2]

- (c) The circuit of Fig. 5.1 is modified so that it produces full-wave rectification of an input voltage. Suggest, with a reason, how V_{OUT} now varies with time when V_{IN} is as shown in Fig. 5.2.

.....

 [2]

22. M/J/23/42/Q7

Four diodes are used in a bridge rectifier circuit to produce rectification of a sinusoidal a.c. input voltage V_{IN} .

Fig. 7.1 shows part of the circuit, but three of the diodes are missing.

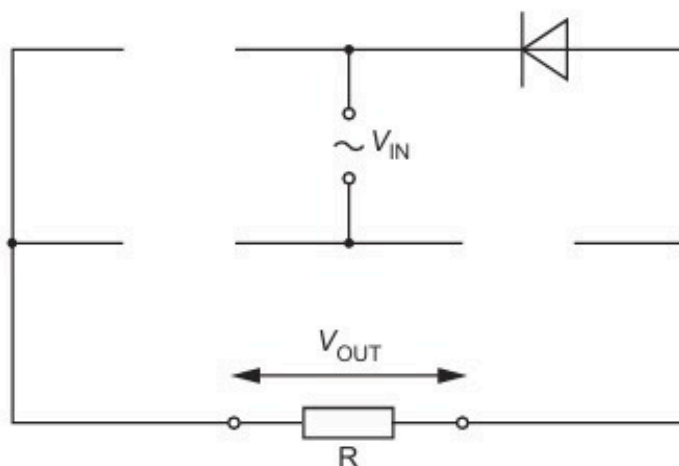


Fig. 7.1

The p.d. across the load resistor R is the output p.d. V_{OUT} of the bridge rectifier.

(a) (i) State the name of the type of rectification produced by a bridge rectifier.

..... [1]

(ii) Complete Fig. 7.1 by drawing the three missing diodes, correctly connected. [2]

(iii) On Fig. 7.1, draw an arrow to indicate the direction of the current in resistor R . [1]

(b) V_{IN} has amplitude V_0 and period T . Fig. 7.2 shows the variation with time t of V_{IN} .

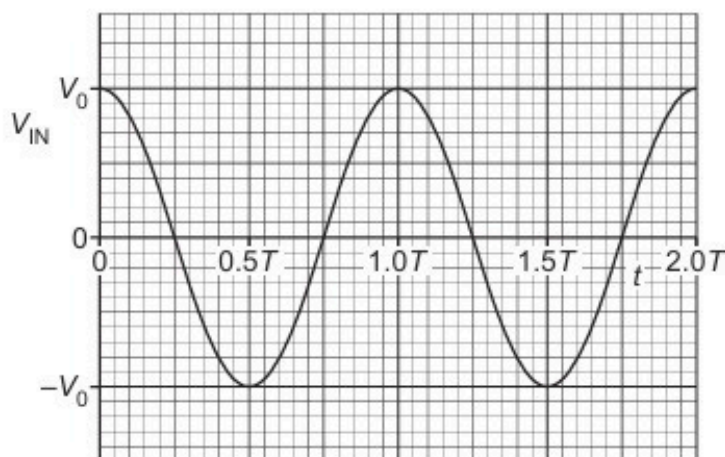


Fig. 7.2

- (i) On Fig. 7.3, sketch the variation of V_{OUT} with t between $t = 0$ and $t = 2.0T$.

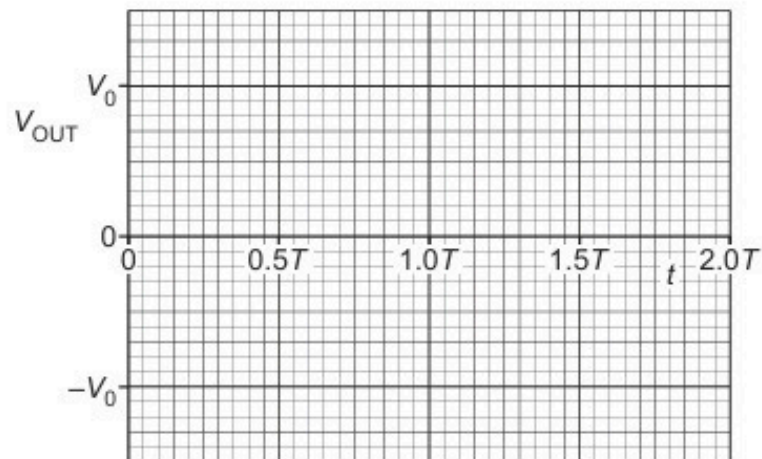


Fig. 7.3

[3]

- (ii) The power dissipated in the resistor is P .

On Fig. 7.4, sketch the variation of P with t between $t = 0$ and $t = 2.0T$.

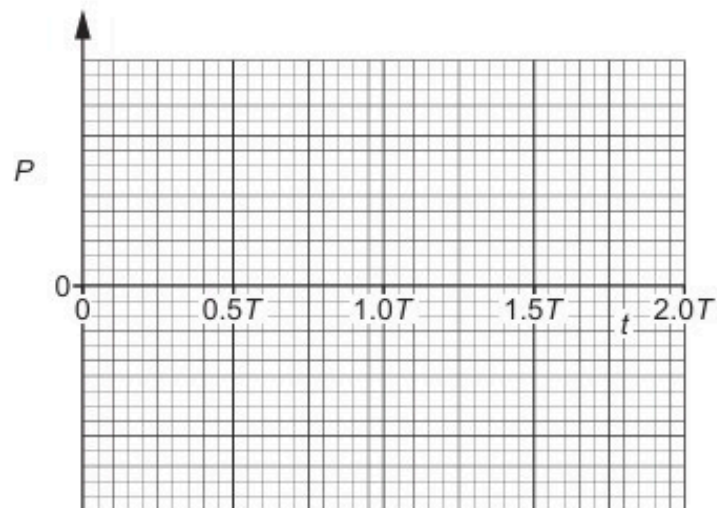


Fig. 7.4

[2]

- (iii) Suggest, with a reason, how the root-mean-square (r.m.s.) value of V_{OUT} compares with the r.m.s. value of V_{IN} .

.....

..... [1]

[Total: 10]

23. O/N/23/41/Q7

A varying current I passes through a resistor of resistance R in the circuit shown in Fig. 7.1.

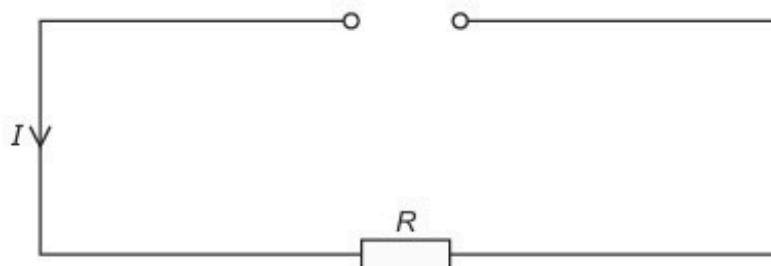


Fig. 7.1

Fig. 7.2 shows the variation with time t of I .

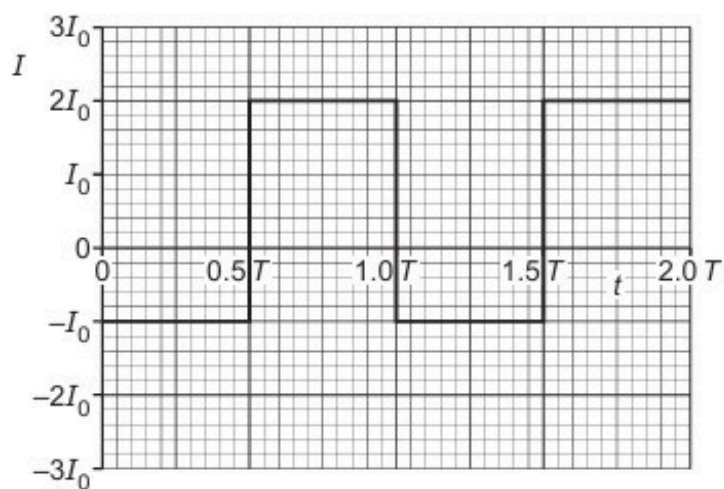


Fig. 7.2

The current has magnitude $2I_0$ when it is in the positive direction and I_0 when it is in the negative direction. The period of the variation of the current is T .

- (a) Determine expressions, in terms of I_0 and R , for the power P dissipated in the resistor for the times when:

- (i) the current is in the negative direction

$P = \dots\dots\dots [1]$

- (ii) the current is in the positive direction.

$P = \dots\dots\dots [1]$

- (b) On Fig. 7.3, sketch the variation of P with t between $t = 0$ and $t = 2.0T$. Label the power axis with an appropriate scale.

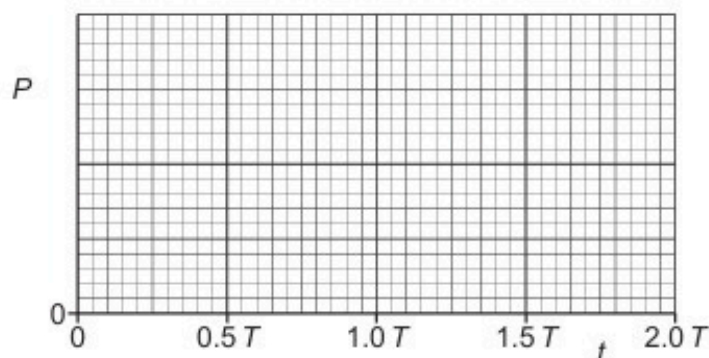


Fig. 7.3

[3]

- (c) Use your answer in (b) to determine an expression, in terms of I_0 and R , for:

- (i) the mean power $\langle P \rangle$ in the resistor

$$\langle P \rangle = \dots\dots\dots [1]$$

- (ii) the root-mean-square (r.m.s.) current $I_{\text{r.m.s.}}$ in the resistor.

$$I_{\text{r.m.s.}} = \dots\dots\dots [2]$$

[Total: 8]

ANSWERS

1.

- (a) e.g. more (output) power available
 e.g. less ripple for same smoothing capacitor
any sensible suggestion B1
- (b) (i) curve showing half-wave rectification B1
 (ii) similar to (i) but phase shift of 180° B1
- (c) (i) correct symbol, connected in parallel with R B1
 (ii) 1 larger capacitor / second capacitor in parallel with R B1
 (not increase R)
 2 same peak values B1
 correct shape giving less ripple B1

2.

- (a) *either* the value of steady / constant voltage M1
 that produces same power (in a resistor) as the alternating voltage A1
or if alternating voltage is squared and averaged (M1)
 the r.m.s. value is the square root of this averaged value (A1)
- (b) (i) 220 V A1
 (ii) 156 V A1
 (iii) 60 Hz A1
- (c) power = V_{rms}^2 / R C1
 $R = 156^2 / 1500$
 $= 16 \Omega$ A1

3.

- (a) power / heating depends on I^2 M1
 so independent of current direction A1
- (b) *either* maximum power = $I_0^2 R$ or average power = $I_{\text{RMS}}^2 R$ M1
 $I_0 = \sqrt{2} \times I_{\text{RMS}}$ M1
 maximum power = $2 \times$ average power
 ratio = 0.5 A1

4.

(a) (i) $2\pi f = 380$ C1
frequency = 60 Hz A1

(ii) $I_{\text{RMS}} \times \sqrt{2} = I_0$ C1
 $I_{\text{RMS}} = 9.9 / \sqrt{2}$
 $= 7.0 \text{ A}$ A1

(b) power = $I^2 R$ C1
 $R = 400 / 7.0^2$
 $= 8.2 \Omega$ A1

5.

(a) (i) period = $1/50$ C1
 $t_1 = 0.03 \text{ s}$ A1

(ii) peak voltage = 17.0 V A1

(iii) r.m.s. voltage = $17.0/\sqrt{2}$
 $= 12.0 \text{ V}$ A1

(iv) mean voltage = 0 A1

(b) power = V^2/R C1
 $= 12^2/2.4$
 $= 60 \text{ W}$ A1

6.

(a) supply connected correctly (to left & right) B1
load connected correctly (to top & bottom) B1

(b) e.g. power supplied on every half-cycle
greater average/mean power
(any sensible suggestion, 1 mark) B1

(c) (i) reduction in the variation of the output voltage/current B1

(ii) larger capacitance produces more smoothing M1
either product RC larger
or for the same load A1

7.

- (a) (i) peak voltage = 4.0 V A1
- (ii) r.m.s. voltage $(= 4.0/\sqrt{2}) = 2.8$ V A1
- (iii) period $T = 20$ ms M1
frequency = $1 / (20 \times 10^{-3})$ M1
frequency = 50 Hz A0
- (b) (i) change = $4.0 - 2.4 = 1.6$ V A1
- (ii) $\Delta Q = C\Delta V$ or $Q = CV$ C1
 $= 5.0 \times 10^{-6} \times 1.6 = 8.0 \times 10^{-6}$ C A1
- (iii) discharge time = 7 ms C1
current = $(8.0 \times 10^{-6}) / (7.0 \times 10^{-3})$ M1
 $= 1.1(4) \times 10^{-3}$ A A0
- (c) average p.d. = 3.2 V C1
resistance = $3.2 / (1.1 \times 10^{-3})$
 $= 2900 \Omega$ (allow 2800 Ω) A1

8.

- (a) (i) connection to 'top' of resistor labelled as positive B1
- (ii) diode B and diode D B1
- (b) (i) $V_P = 4.0$ V C1
mean power = $V_P^2/2R$ C1
 $= 4^2 / (2 \times 2700)$
 $= 2.96 \times 10^{-3}$ W A1
- (ii) capacitor, correct symbol, connected in parallel with R B1
- (c) graph: half-wave rectification M1
same period and same peak value A1

9.

(a) (i) $V_0 (= 14\sqrt{2}) = 19.8 \text{ (20) V}$ A1

(ii) $\omega (= 2\pi \times 750) = 4700 \text{ rad s}^{-1}$ A1

(b) large amount of charge required to charge capacitor M1
capacitor would charge and discharge rapidly/in a very short time
or capacitor would charge and discharge 750/1500 times per second M1

$I = Q/t$, so large current A1

10.

(a) period = 15 ms C1

frequency $(= 1/T) = 67 \text{ Hz}$ A1

(b) zero A1

(c) $I_{\text{r.m.s.}} = I_0 / \sqrt{2}$ C1

$= 0.53 \text{ A}$ A1

(d) energy $= I_{\text{r.m.s.}}^2 \times R \times t$ or $\frac{1}{2} I_0^2 \times R \times t$
or
power $= I_{\text{r.m.s.}}^2 \times R$ and energy $= \text{power} \times t$ C1

energy $= 0.53^2 \times 450 \times 30 \times 10^{-3}$
 $= 3.8 \text{ J}$ A1

11.

(a) point P shown at 'lower end' of load B1

(b) $V_{\text{r.m.s.}} = 6.0 / \sqrt{2} = 4.24 \text{ V}$ C1

$I_{\text{r.m.s.}} = 4.24 / (2.4 \times 10^3)$
 $= 1.8 \times 10^{-3} \text{ A}$ A1

(c) (i)	capacitor in parallel with load	B1
(ii)	line from peak to curve at 3.0 V for either half- or full-wave rectified	M1
	correct curvature on line (gradient becoming more shallow)	A1
	line drawn as for full-wave rectified	A1

12.

11(a)(i)	circles drawn only around the top left and bottom right diodes	B1
11(a)(ii)	B shown as (+)ve and A shown as (–)ve	B1
11(b)(i)	$V_{r.m.s.} (= 5.6 / \sqrt{2}) = 4.0 \text{ V}$	A1
11(b)(ii)	$380 = 2\pi f$ or $f = 60.5 \text{ Hz}$	C1
	number $(= 2f) = 120$	A1
11(c)(i)	peak values (all) unchanged	B1
	(all) minima shown at 4.0 V	B1
	three lines from near peak showing concave curves after leaving dotted line not 'kinked' and not cutting the peak reaching <u>candidate's</u> minimum at the point where the decay meets the next dotted line	B1
	three lines drawn along the dotted lines showing rise in voltage from minima back to peak values	B1
11(c)(ii)	<u>mean</u> p.d. is higher or <u>r.m.s.</u> p.d. is higher or capacitor supplies energy <u>to resistor</u>	M1
	so (mean) power increases	A1

13.

10(a)	$V_{MAX} = 15 \text{ V}$	A1
10(b)	$210 = 2\pi / T$	C1
	$T = 0.0299 \text{ s}$	C1
	$(t_2 - t_1) = 0.060 \text{ s}$	A1

14.

10(a)(i)	lower right and upper left diodes circled	B1
10(a)(ii)	maximum $= 7.0\sqrt{2}$	A1
	$= 9.9 \text{ V}$	
10(b)(i)	correct symbol for capacitor, shown connected in parallel with R	B1
10(b)(ii)	1. (ripple) decreases	B1
	2. (ripple) increases	B1

15.

10(a)	230 V	A1
10(b)	$\omega = 100\pi$	C1
	$T = \frac{2\pi}{\omega} = \frac{2\pi}{100\pi}$ $= 0.020 \text{ s}$	A1
10(c)(i)	half-wave (rectification)	B1
10(c)(ii)	sinusoidal half waves in positive V only or negative V only, peak at 320 V	B1
	line at zero for second half of cycle	B1
	two time periods shown, each of 0.020 s	B1
10(c)(iii)	capacitor added in parallel with resistor	B1

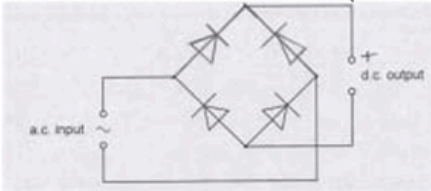
16.

10(a)	the steady current or the direct current	M1
	that produces the same heating effect (as the alternating current)	A1
10(b)(i)	peak current = 2.6 A and r.m.s. current = 1.8 A	A1
10(b)(ii)	peak current = 2.0 A and r.m.s. current = 2.0 A	A1
10(c)(i)	$k = 2\pi f$	C1
	$= 2\pi \times 50$	A1
	$= 310 \text{ rad s}^{-1}$	
10(c)(ii)	power = V_{RMS}^2 / R or power = $V_0^2 / 2R$	C1
	$R = (240 / \sqrt{2})^2 / 3200$ or $R = 240^2 / (2 \times 3200)$	A1
	$R = 9.0 \Omega$	

17.

9(a)	constant voltage	M1
	that produces/dissipates same power as (the mean power of) the alternating voltage	A1
9(b)(i)	(maximum) rate of cutting of (magnetic) flux doubles	B1
	(peak and hence) r.m.s. induced e.m.f. doubles	B1
9(b)(ii)	sketch: (sinusoidal) wave of period 10 ms	B1
	peak E shown as $\pm 34 \text{ V}$	B2
	(1 mark out of 2 awarded if peak E shown as $\pm 17 \text{ V}$ or $\pm 24 \text{ V}$)	
9(c)	current in the coil results in forces that oppose its rotation or current in the resistor dissipates the energy of rotation	B1
	coil stops rotating	B1

18.

7(a)(i)	two diodes added in correct directions (Both diodes pointing inwards and upwards), correct symbols only 	B1
7(a)(ii)	'+' anywhere on upper output wire	B1
7(b)(i)	$\omega = 2\pi / T$ $= 2\pi / 2.5$ $= 0.80 \pi$ or $4\pi / 5$ or 2.5	C1
	(V =) 3.5 sin (0.8π t) or $3.5 \sin (4\pi t / 5)$ or $3.5 \sin (2.5 t)$	A1
7(b)(ii)	$(P =) \frac{V^2}{2R}$ or $(P =) \frac{V_{r.m.s.}^2}{R}$ $= \frac{3.5^2}{2 \times 12}$ or $\frac{2.47^2}{12}$	C1
	= 0.51 W	A1

19.

5(a)(i)	conversion (from a.c.) to d.c.	B1
5(a)(ii)	full-wave (rectification)	B1
5(b)(i)	P labelled – and Q labelled +	B1
5(b)(ii)	V_{OUT} scale labelled 4 and 8 on the 2 cm tick marks	B1
	$T = 2\pi / \omega$ $= 2\pi / 25\pi$ $= 0.08 \text{ s}$	C1
	t scale labelled 0.02, 0.04, 0.06, 0.08, 0.10, 0.12 on the 2 cm tick marks	A1
5(c)(i)	correct symbol used for capacitor and capacitor connected in parallel with the 1.2 kΩ resistor.	B1
5(c)(ii)	straight lines or curves, with negative decreasing gradients, drawn between adjacent peaks, from top of first peak to meet line going up to next peak	B1
	lines, from one peak to the line going up to the next peak, show a drop in p.d. of 1½ small squares	B1
5(c)(iii)	$V = 0.90 \times 6.0 (= 5.4 \text{ V})$ or discharge time (for each cycle) = 0.034 s	C1
	$V = V_0 \exp (-t / RC)$ $5.4 = 6.0 \exp [-0.034 / (1.2 \times 10^3 \times C)]$	C1
	$C = 2.7 \times 10^{-4} \text{ F}$	A1

20.

7(a)(i)	peak voltage = $4.2 \times \sqrt{2}$ (= 5.9 V)	B1
	power = V^2 / R = $5.9^2 / 760 = 0.046 \text{ W}$ or 46 mW	A1
7(a)(ii)	sketch shows peak(s) in power at 46 mW	B1
	correct shape (sinusoidal wave sitting on t -axis)	B1
	four cycles of repeating pattern shown, with $P = 0$ at 0, 10, 20, 30, 40 μs	B1
7(a)(iii)	line is symmetrical about 23 mW	B1
7(b)(i)	(alternating p.d. makes) the crystal vibrate	B1
	vibrations (of crystal) causes air to vibrate	B1
	frequency is in ultrasound range	B1
7(b)(ii)	(air makes) crystal vibrate, which causes an e.m.f. to be generated across the (second) crystal	B1

21.

5(a)(i)	correct circuit symbol for a diode shown correctly connected in series with the wires leading into and out of the dotted box	B1
5(a)(ii)	smoothing / V_{OUT} is smoothed	B1
5(b)(i)	frequency = $1 / 0.04$ = 25 Hz	A1
5(b)(ii)	$V = V_0 \exp(-t / RC)$ and $\tau = RC$ or $V = V_0 \exp(-t / \tau)$	C1
	$3.25 = 5.50 \exp(-0.020 / \tau)$ leading to $\tau = 0.038 \text{ s}$	A1
5(b)(iii)	$\tau = RC$	C1
	capacitance = $0.038 / 14000$ = $2.7 \times 10^{-6} \text{ F}$	A1
5(c)	V_{IN} has constant magnitude in both positive and negative directions	B1
	(so) V_{OUT} is (now) constant / V_{OUT} does not vary with time	B1

22.

7(a)(i)	full-wave (rectification)	B1
7(a)(ii)	lower left diode shown pointing left	B1
	lower right and upper left diodes shown pointing left	B1
7(a)(iii)	arrow indicating current direction in resistor to the right	B1
7(b)(i)	sketch: periodic line showing minimum $V_{\text{OUT}} = 0$ and maximum $V_{\text{OUT}} = +V_0$	B1
	line showing peak V_{OUT} at $t = 0, 0.5T, 1.0T, 1.5T$ and $2.0T$, with V_{OUT} going to zero half-way in between each peak	B1
	line showing correct modulated sine shape	B1
7(b)(ii)	sketch: sinusoidal curve with troughs sitting on the time axis	B1
	peak power at $t = 0, 0.5T, 1.0T, 1.5T$ and $2.0T$ and zero power half-way in between each peak	B1
7(b)(iii)	same power-time graph with or without rectification, so same V_{rms} or V^2 -time graph is same for both V_{OUT} and V_{IN} , so same V_{rms} or power does not depend on sign of V , so same V_{rms}	B1

23.

7(a)(i)	$P = I_0^2 R$	A1
7(a)(ii)	$P = 4I_0^2 R$	A1
7(b)	sketch: square wave of period T , with P always non-zero	B1
	horizontal lines, from 0 to $0.5T$ and from $1.0T$ to $1.5T$, all at the same level that the scale indicates to be $I_0^2 R$	B1
	horizontal lines, from $0.5T$ to $1.0T$ and from $1.5T$ to $2.0T$, at a level that is four times higher than the lower lines	B1
7(c)(i)	$\langle P \rangle = (5/2)I_0^2 R$	A1
7(c)(ii)	$\langle P \rangle = I_{r.m.s.}^2 R$	C1
	$I_{r.m.s.}^2 R = (5/2)I_0^2 R$	A1
	$I_{r.m.s.} = \sqrt{(5/2)} I_0$	